

### 5.5.1 Integration by Substitution (Indefinite Integrals)

**Motivation:** Suppose we wanted to determine  $\int 2x\sqrt{1+x^2} dx$ .

**Terminology:** We call this approach of “reversing” the Chain Rule a ***u*-substitution**.

**Theorem:** Suppose  $g$  is a differentiable function whose range is an  $I$ ,  $f$  is continuous on  $I$ , and  $u = g(x) \Rightarrow du = g'(x)dx$ , then

$$\int f(g(x)) \cdot g'(x) dx = \int f(u) du$$

*Note:* In the motivation above we had  $f(x) = \sqrt{x}$  and  $g(x) = 1 + x^2$ .

*Note:* The most difficult part of the  $u$ -substitution process is determining what expression to let  $u$  equal. Unfortunately, there is no general algorithm for determining the  $u$ , but one can get a feel for what to let  $u$  equal by experiencing many problems.

**Example 1:** Find  $\int x^3 \cos(x^4 + 2) dx$

**Example 2:** Find  $\int \frac{x}{\sqrt{1-4x^2}} dx$

**Example 3:** Find  $\int \tan(x) dx$

**Example 4:** Find  $\int x^5 \sqrt{1+x^2} dx$ .

**Example 5:** Determine each of the following.

a) Find  $\int \frac{\sin(\sqrt{x})}{\sqrt{x}} dx$ .

b) Find  $\int e^{-5r} dr$ .

c) Find  $\int \cot(x) dx$ .

d) Find  $\int \frac{2^t}{2^t + 3} dt$ .

e) Find  $\int \frac{x}{1+x^4} dx$ .

f) Find  $\int x^{13} (x^7 + 9)^{19} dx$